THEOREM 2.13. *Suppose M is a closed, oriented, connected 3-manifold with $\beta_1(M) = 1$ and $\phi : \pi_1(M) \longrightarrow \Gamma$ a nontrivial* PTFA *coefficient system. Suppose $\mathcal{R}$ is a ring such that $\mathbb{Z}\Gamma \subseteq \mathcal{R} \subseteq \mathcal{K}$. Then there is a symmetric linking form*

$$B\ell : H_1(M;\mathcal{R}) \longrightarrow H_1(M;\mathcal{R})^{\#}$$

defined on the higher-order Alexander module $\mathcal{A} := H_1(M;\mathcal{R})$. *If either $\mathcal{R}$ is a* PID, *or some element of the augmentation ideal of $\mathbb{Z}\pi_1(M)$ is sent (under ϕ) to an invertible element of $\mathcal{R}$, then $B\ell$ is nonsingular.*

Proof. Note that $\mathcal{A}$ is a torsion $\mathcal{R}$-module by Proposition 2.11, since $\mathcal{K}$ is also the quotient field of the Ore domain $\mathcal{R}$. Define $B\ell$ as the composition of the Poincaré duality isomorphism to $H^2(M;\mathcal{R})$, the inverse of the Bockstein to $H^1(M;\mathcal{K}/\mathcal{R})$, and the usual Kronecker evaluation map to $\mathcal{A}^{\#}$. The Bockstein

$$B : H^1(M;\mathcal{K}/\mathcal{R}) \longrightarrow H^2(M;\mathcal{R})$$

associated to the short exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\mathcal{R} \longrightarrow 0$$

is an isomorphism since $H^2(M;\mathcal{K}) \cong H_1(M;\mathcal{K}) = 0$ by Proposition 2.11, and $H^1(M;\mathcal{K}) = 0$ by Proposition 2.11 and Remark 2.8.3. Under the second hypothesis on $\mathcal{R}$, the Kronecker evaluation map

$$H^1(M;\mathcal{K}/\mathcal{R}) \longrightarrow \mathrm{Hom}_{\mathcal{R}}\left(H_1(M;\mathcal{R}), \mathcal{K}/\mathcal{R}\right)$$

is an isomorphism by the UCSS since $H_0(M;\mathcal{R}) = 0$ (see Remark 2.8.3 and Proposition 2.9). If $\mathcal{R}$ is a PID then $\mathcal{K}/\mathcal{R}$ is an injective $\mathcal{R}$-module since it is clearly divisible [Ste, I Prop. 6.10]. Thus

$$\mathrm{Ext}^i_{\mathcal{R}}\left(H_0(M;\mathcal{R}), \mathcal{K}/\mathcal{R}\right) = 0$$

for $i > 0$ and therefore the Kronecker map is an isomorphism.

We need to show that $\mathcal{A}$ has homological dimension one and is finitely generated. This is immediate if $\mathcal{R}$ is a PID [Ste, p. 22]. Since, in this paper we shall only need this special case we omit the proof of the general case.

We also need to show that $B\ell$ is "conjugate symmetric". The diagram below commutes up to a sign (see, for example, [M, p. 410]), where B' is the homology Bockstein

$$\begin{array}{ccc}
H_2(M;\mathcal{K}/\mathcal{R}) & \xrightarrow{B'} & H_1(M;\mathcal{R}) \\
\cong\big\downarrow \text{P.D.} & & \cong\big\downarrow \text{P.D.} \\
H^1(M;\mathcal{K}/\mathcal{R}) & \xrightarrow{B} & H^2(M;\mathcal{R}) \\
\big\downarrow \kappa & & \\
\mathrm{Hom}_{\mathcal{R}}(H_1(M;\mathcal{R}), \mathcal{K}/\mathcal{R}) & &
\end{array}$$